UAV Health Monitoring

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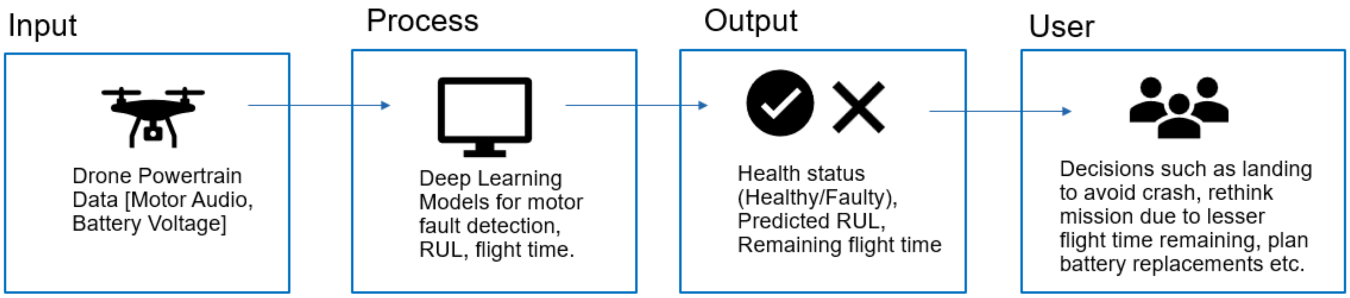
User Stories

1. *As a UAV operator, I want to automatically detect motor or propeller faults during flight, so that I can ensure safe landing and prevent crashes in mission-critical operations.*
2. *As a drone fleet manager, I want to track battery health and predict remaining flight time, so that I can plan missions more reliably and reduce unexpected downtime.*
3. *As a remote operator using a drone-in-a-box system, I want in-flight monitoring of propeller damage, so that I can identify issues that are not visually observable and avoid risky deployments.*
4. *As a UAV maintenance technician, I want pre-flight diagnostics to flag potential motor or propeller issues, so that I can repair or replace components before they fail in operation.*
5. *As an aviation safety regulator, I want a standardized health monitoring framework integrated into UAVs, so that I can improve operational safety and ensure compliance with safety management systems.*

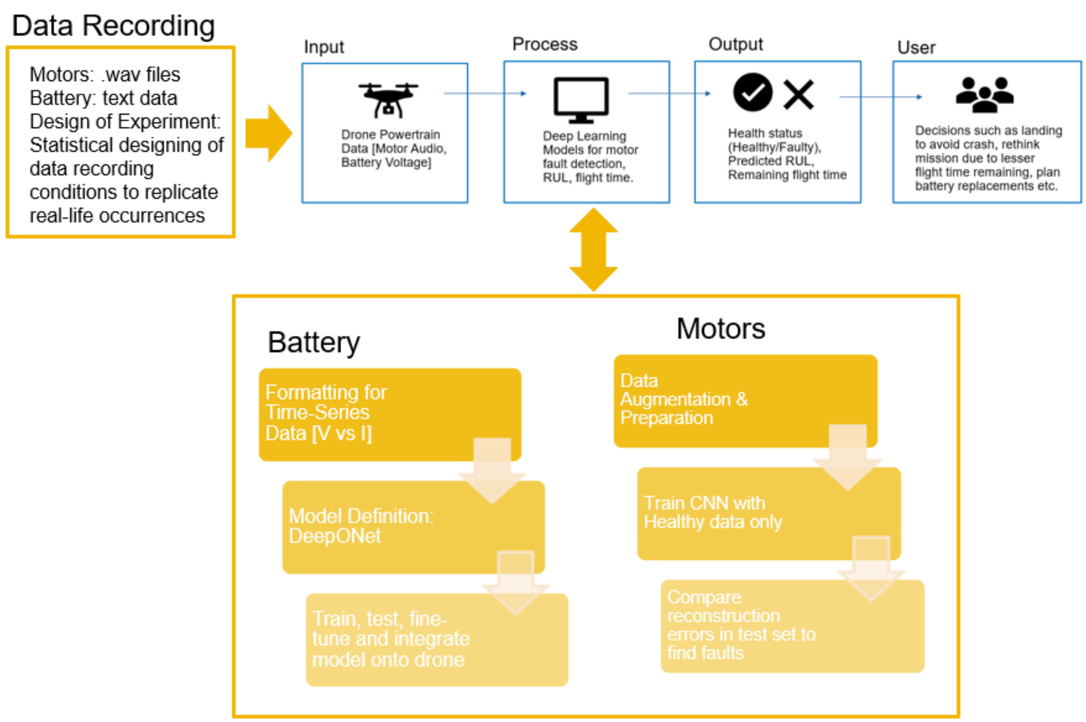
Design Conventions:

- Boxes represent a set of information, labelled where necessary such as input, output etc.
- Arrows represent transition from one box to the next, i.e., from one set of steps to the next. $\Longrightarrow$
- 2-sided arrows represent an extension of information provided by 1 box, in another box. $\Longleftrightarrow$
- Blue boxes and arrows represent the highest-level view of the design.
- Additional yellow boxes and arrows represent a lower-level view.
- Green boxes and arrows represent the lowest level view of the design.

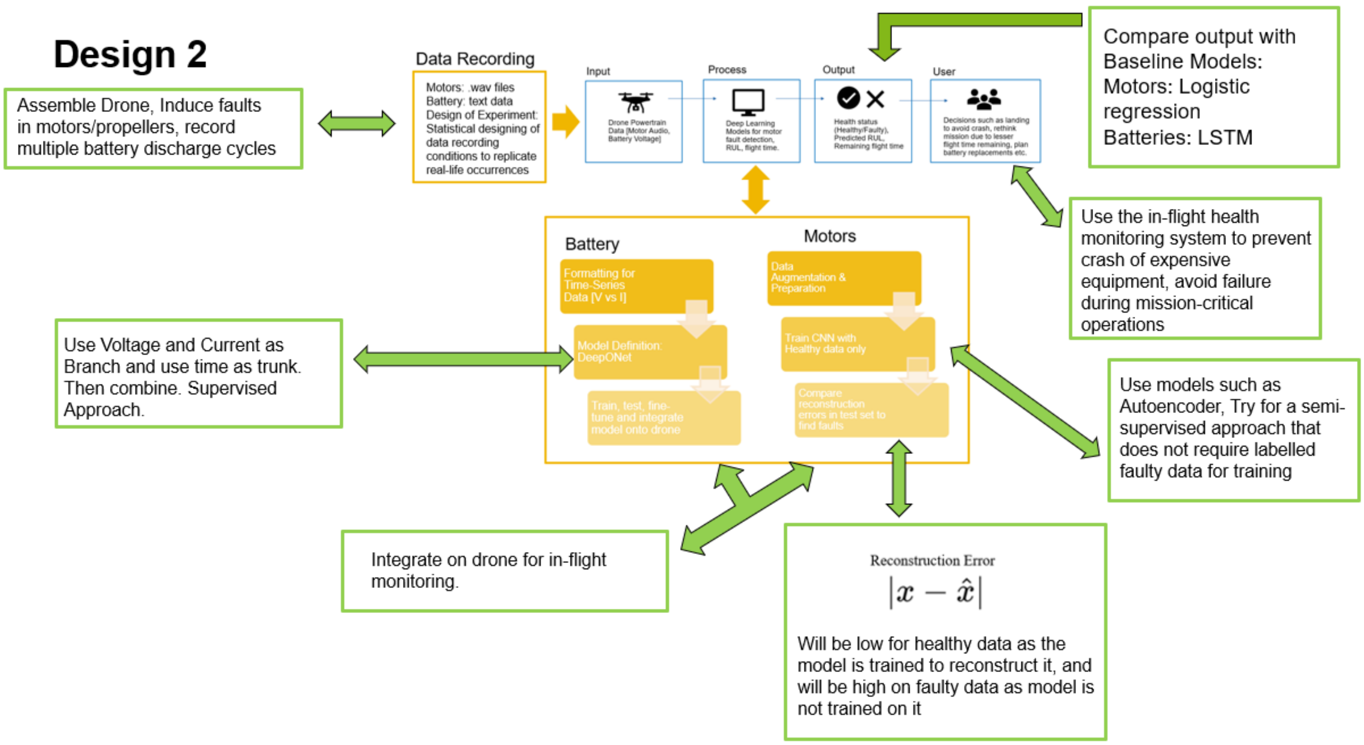
Design 0



Design 1



Design 2



Assemble Drone, Induce faults in motors/propellers, record multiple battery discharge cycles
Data Recording
Motors: .wav files
Battery: text data
Design of Experiment: Statistical designing of data recording conditions to replicate real-life occurrences
Input
Drone Powertrain Data [Motor Audio, Battery Voltage]
Process
Deep Learning Models for motor fault detection, RUL, flight time.
Output
Health status (Healthy/Faulty), Predicted RUL, Remaining flight time
User
Decisions such as landing to avoid crash, rethink mission due to lesser flight time remaining, plan battery replacements etc.
Compare output with Baseline Models:
Motors: Logistic regression
Batteries: LSTM
Use the in-flight health monitoring system to prevent crash of expensive equipment, avoid failure during mission-critical operations
Battery
Formatting for Time-Series Data [V vs I]
Model Definition: DeepONet
Train, test, fine-tune and integrate model onto drone
Motors
Data Augmentation & Preparation
Train CNN with Healthy data only
Compare reconstruction errors in test set to find faults
Use Voltage and Current as Branch and use time as trunk. Then combine. Supervised Approach.
Use models such as Autoencoder, Try for a semi-supervised approach that does not require labelled faulty data for training
Integrate on drone for in-flight monitoring.
Reconstruction Error
$|x - \hat{x}|$
Will be low for healthy data as the model is trained to reconstruct it, and will be high on faulty data as model is not trained on it